CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

April 2013

CL 210 Hydrology & Hydraulics

Date: 25.04.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1(a) Explain: Evaporation. Also, discuss the measure for the control of evaporation from a large water body. [03]

(b) A storm with 12.0 cm precipitation produced a direct runoff of 6.8 cm. Given the time distribution of the storm as below, estimate the Φ index of the storm. [04]

Time from start (hr)	1	2	3	4	5	6	7	8
Incremental rainfall in each hour (cm)	0.5	1.3	2	3	1.9	1.7	1.1	0.5

Q – 2(a) Distinguish between aquitard, aquiclude and aquifuge. [7]

In an artesian aquifer of 8 m thick, a 10 cm diameter well is pumped at a constant rate of 100 lit/minute. The steady state drawdown observed in two wells located at 10 m and 50 m distances from the centre of the well are 3 m and 0.05 m respectively. Compute the transmissibility and the hydraulic conductivity of the aquifer.

(b) Explain the following terms: [4]

(i) Storage coefficient (ii) specific yield (iii) Hydraulic conductivity (iv) Aquifer

(c) List the assumptions made in the analysis of steady radial flow into well. [3]

OR

Q – 2(a) Derive an expression for the steady state discharge of well fully penetrating into a confined aquifer. [7]

(b) Enlist various sub-surface techniques of ground water investigation. Discuss any two techniques in detail. [7]

Q – 3(a) Explain the necessity of groundwater recharge. Discus is the natural phenomenon of groundwater recharge. [7]

(b) Discuss the phenomenon of salt water intrusion. What are its causes and effects on the water supply system? [7]

OR

Q – 3(a) Enlist various artificial methods of groundwater recharge. Explain, in detail, any TWO of them. [7]

(b) Discuss, with a neat sketch, Ghyben-Herzberg relation for salt water intrusion. How can it be controlled? [7]

SECTION – II

Q – 4 (a) Discuss briefly various factors affecting runoff. [03]

(b) Flood-frequency computations for the river chambal at Gandhisagar dam, by using Gumbel's method, yielded the following results: [04]

Return Period T (years)	Peak flood (m^3/s)
50	40,809
100	46,300

Estimate the flood magnitude in this river with a return period of 500 years.

Q - 5 (a) Define: Conjunctive Use. What are the needs of conjunctive use? [04]

(b) A catchment area has 8 raingauge stations. In a year the annual rainfall recorded by raingauges are as follows [04]

Station	A	B	C	D	E	F	G	H
Rainfall (cm)	140	135	120	115	150	110	160	150

For an error of 8% in the estimation of mean rainfall, calculate the minimum number of additional station required to be established in the catchment area.

(c) It was observed in a field test that 2 hr 50 min was required for a tracer to travel from one well to another 18 m apart and the difference in their water surface elevations was 0.75 m. Samples of the aquifer between the wells indicated a porosity of 10%. Determine the permeability of the aquifer, seepage velocity and the Reynolds number for the flow, assuming an average grain size of 1 mm and ν at 27°C = 0.009 Stoke. [06]

OR

Q - 5 (a) Explain the laboratory methods of measurement of permeability. [04]

(b) Given below are the ordinate of a 6-hr unit hydrograph for a catchment. Calculate the ordinates of the DRH due to a rainfall excess of 3.5 cm occurring in 6 hr. [04]

Time (hr)	0	3	6	9	12	15	18	24	30
UH ordinate (m^3/s)	0	25	50	85	125	160	185	160	110
Time (hr)	36	42	48	54	60	69			
UH ordinate (m^3/s)	60	36	25	16	8	0			

(c) An aquifer of aerial extent of 120 km^2 is overlain by five strata as given below: [06]

Strata	Thickness (m)	Kx (m/day)	Ky (m/day)
1 (Top)	2	2	0.31
2	3	1.5	0.27
3	1	2.3	0.045
4	4	0.22	0.09
5	2	0.036	0.021

(i) If a 3.5 hr storm occurs producing a total rainfall 150 mm, estimate the recharge into the aquifer, assuming that the piezometric surface in the aquifer is at the bottom of layer 5th and that all layers are saturated.

(ii) If the five layers are underlain by an impermeable strata instead of the aquifer, estimate the lateral flow per unit width through layers, assuming that the layer dip by 0.15 %.

Q - 6 (a) Define: Aquiclude, Aquifuge [02]

(b) What is ground water pollution? Discuss the pollution mechanism briefly. [06]

(c) Briefly discuss the scenario of ground water potential in India. [06]

OR

Q - 6 (a) Define: Transmissibility, Permeability [02]

(b) Explain the 'water conservation'. Which are the techniques to conserve the surface runoff? Discuss in detail. [06]

(c) Discuss the sources and nature of pollution from industrial, agricultural, sewage and nuclear waste. [06]

****